# JC2066: IT Professionals: Ethical, Legal and Social Issues

L02: Machine Learning Fairness

Jan 23rd, 2026

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Department of Computer Science

Where are the rest of the resources for this class?

On Canvas: https://canvas.cityu.edu.hk/courses/68163

# Today’s Agenda

1. **Risks of AI/ML in real world**
2. AI perceptions, explainability and interpretability
3. Investigating bias
4. Ameliorating bias

# Uncertainty & Unpredictability for users

- Relinquishing control to an AI/ML agent can be helpful, but can be much harder to correct or understand if **things go wrong**
- “**Unpredictability**” can be joyful in one kind of experience, and a terrible idea in another

# *Severe* Failure

Self-driving cars crash because of human driver's unexpected behaviors

# *Severe* Failure

**"With enough data and computing power, software can be trained to categorize images or transcribe speech to a high level of accuracy. But it can't easily go beyond the experience of that training. And even the very best algorithms lack the ability to use common sense, or abstract concepts, to refine their interpretation of the world as humans do." -** Wired article (shown →)

TOM SIMONITE BUSINESS 01.11.18 07:00 AM

## WHEN IT COMES TO GORILLAS, GOOGLE PHOTOS REMAINS BLIND

In WIRED's tests, Google Photos did identify some primates, but no gorillas like this one were to be found. RICK MADONIK/TORONTO STAR/GETTY IMAGES

IN 2015, A black software developer embarrassed Google by tweeting that the company's Photos service had labeled photos of him with a black friend as "gorillas." Google declared itself "appalled and genuinely sorry." An engineer who became the public face of the clean-up operation said the label gorilla would no longer be applied to groups of

# *Severe* Failure

XiaoIce spoke foul language and Tay became racist

# *Severe* Failure

**2017 Tay** used some black-listing of ‘bad words’ but could make no moral judgements.

**2018 Zoe** uses both **black-listing** of ‘**bad words**’ and makes **moral judgements**.

# Uncertainty & Unpredictability for companies & designers

- It can be very hard (even impossible today) to predict all the kinds of scenarios the system could wind up in
- The **unkown-unkown** situation: the system won't handle the situation correctly *but won't know it's wrong*
- Model performance will likely change/degrade as time passes
  - Because users *change* how they interact with the system
  - Because it sees **new data** that looks less like it's training data

## **ML/AI error:** Poor model performance

- **Data is expensive** to collect. Prioritizing *what* specific data to collect is essential for companies with limited resources
- Usually solvable by acquiring more training data for the situations the model is weakest at
- Designers can use **rule** or **non-ML based fallbacks** to still deliver the user *some* value when model performance is not good enough for some cases

## **ML/AI error:** Low confidence or *false* High confidence

- Low confidence predictions: the model has lower performance, or the phenomena itself is less predictable
- High confidence (**when the model is really wrong**) is worse
  - Unkown-unkown errors
- Need to give the user some **error correction** or **feedback** method to deal when this happens
- Communicating with the user or providing good **non-AI/ML fallbacks** is key

# **ML/AI error:** Multiple users and input that looks the same to the system

1. Use Alexa to play 1980s favorite pop songs of your parents
2. Use Alexa to play your favorite study music
3. Use Alexa to listen to rock-n-roll with your friends

## What music should Alexa recommend this account play?

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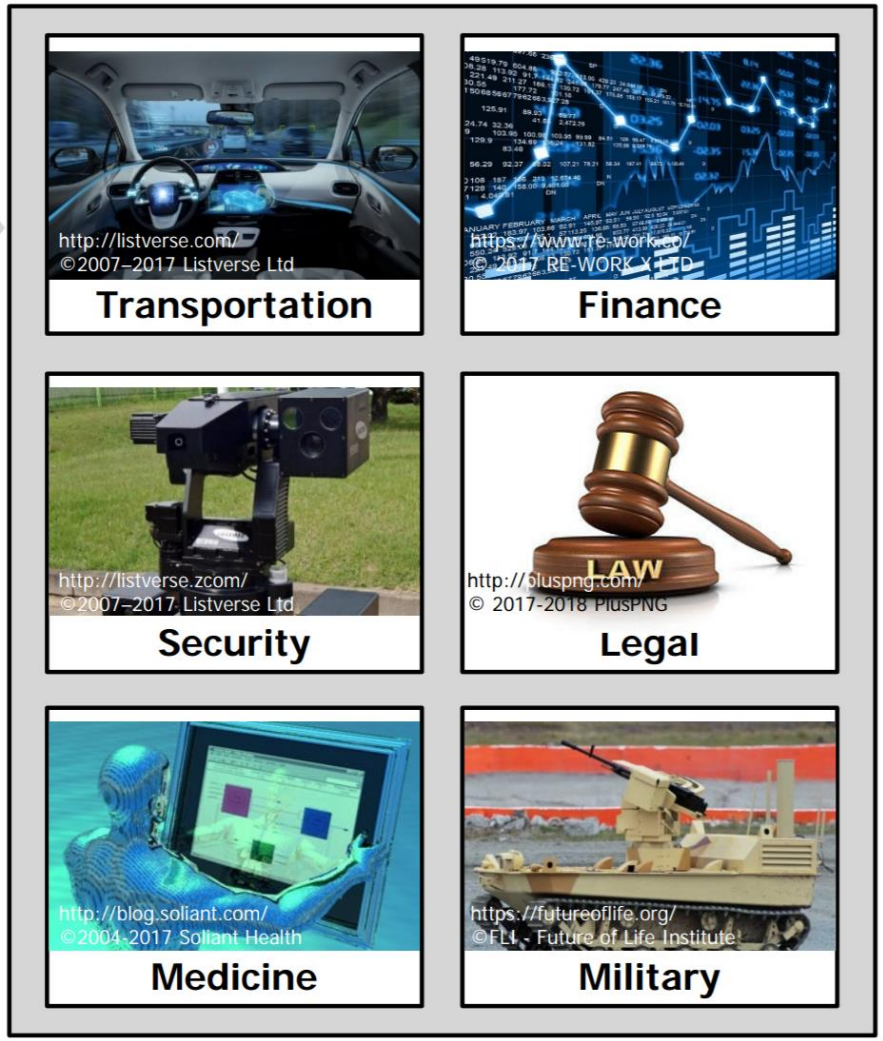
# How do users perceive AI/ML models?




Why did you do that?

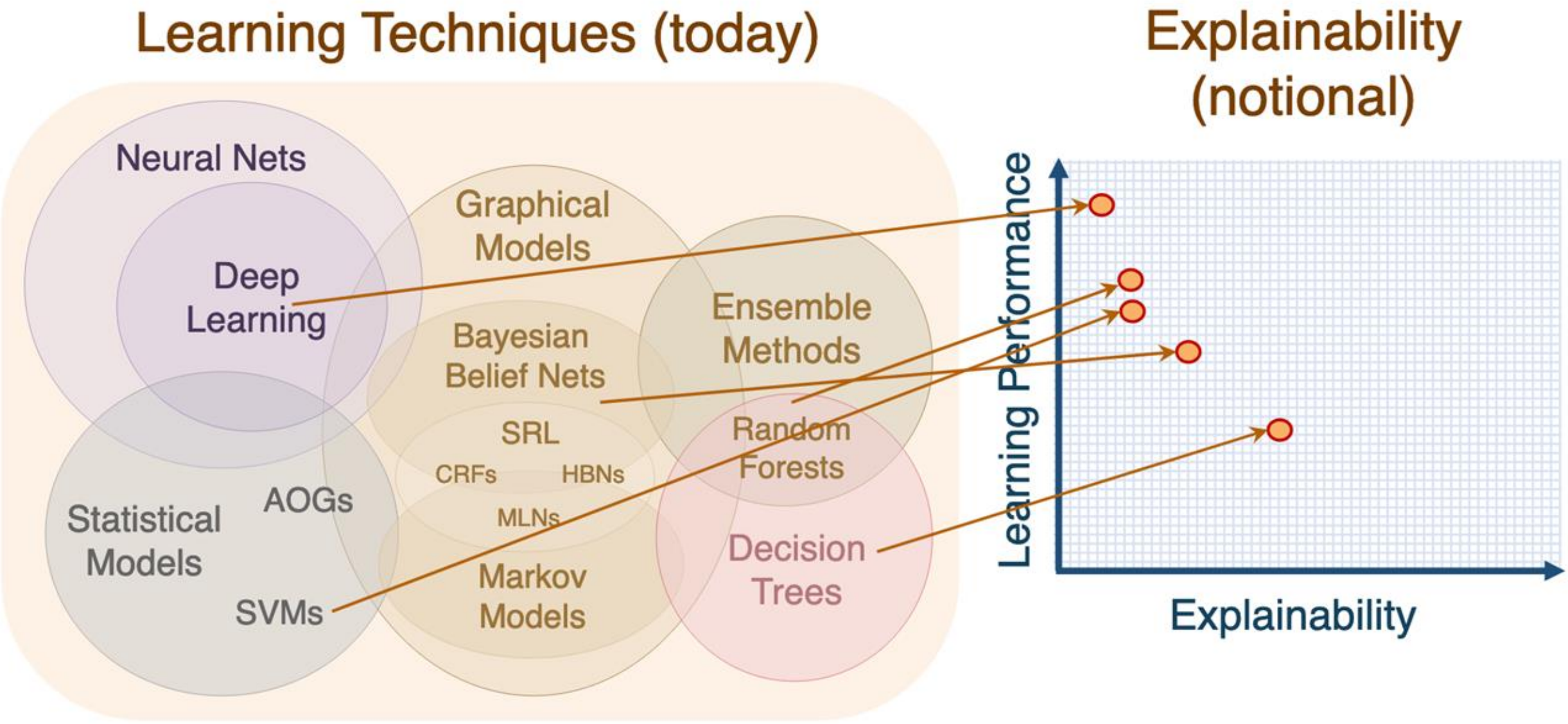
Why not something else?

When do you succeed?

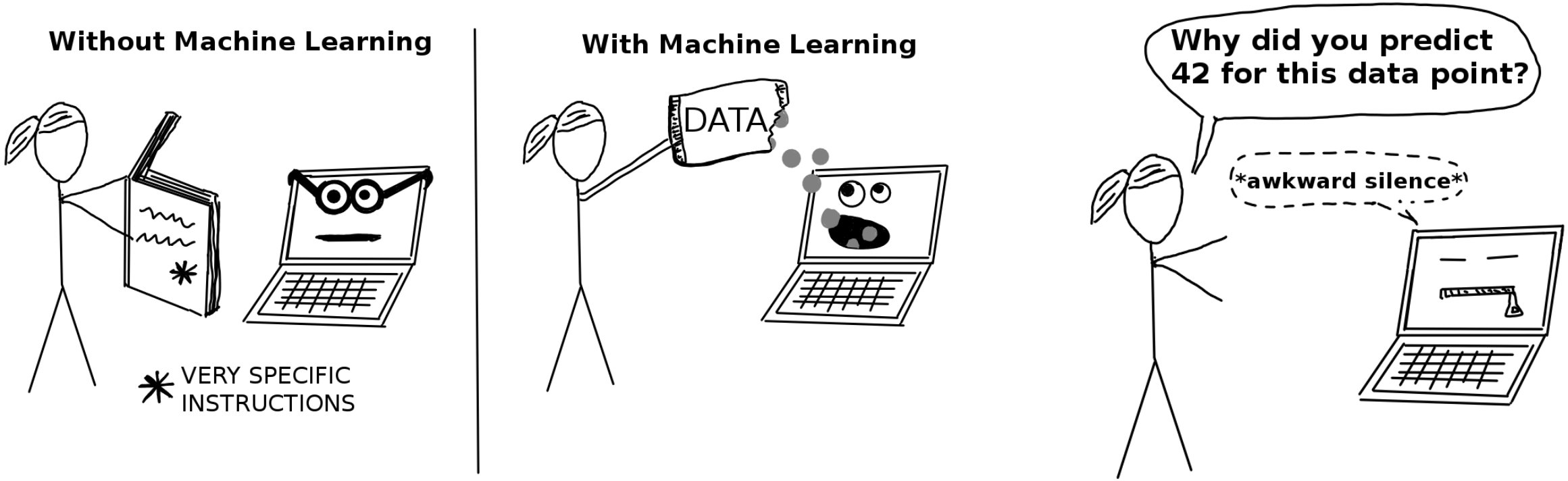
When do you fail?

When can I trust you?

How do I correct an error?


Learning Techniques (today)
Neural Nets
Deep Learning
Graphical Models
Bayesian Belief Nets
Ensemble Methods
SRL
CRFs
HBNs
MLNs
Random Forests
Statistical Models
AOGs
SVMs
Markov Models
Decision Trees
Explainability (notional)
Learning Performance
Explainability

# The Black-box Problem




https://christophm.github.io/interpretable-ml-book/index.html

# Clinicians and ML experts have different needs for explanation ability

ML experts:

- Model transparency
- Post-hoc interpretation

Clinicians:

- No need for “model transparency”
- Clinical assessment as the gold standard
- Ecological validity

# Clinicians and ML experts have different needs for explanation ability

*"We are used to balck-box tools whose processes are not interpretable to us. As long as the performance was always reliable, satisfying, and consistent with the* ***real clinical outcome, we would like to adopt them even if it is difficult for us to make sense of the process****"*

# Clinicians regard ML as assistant tools, and have high expectations

*"Machine Learning is just like another tool that can **assist** us in diagnosing"*

*"Only if the machine performs **significantly better** than average doctors will it be considered useful and helpful. "*

# AI Use Scenarios

AI probably better

- **Recommending** different content to different users
- **Prediction** of future events
- **Personalization**
- **Natural language interactions**
- An agent or **bot**
- UX does not rely on predictability

AI probably **not** better

- The **cost of errors** is very high and overweighs the benefits of a small increase in performance
- Users, customers, or developers need to understand exactly **everything that happens** in the code
- People explicitly convey that they **do not** want a task to be automated or augmented

# Today's Agenda

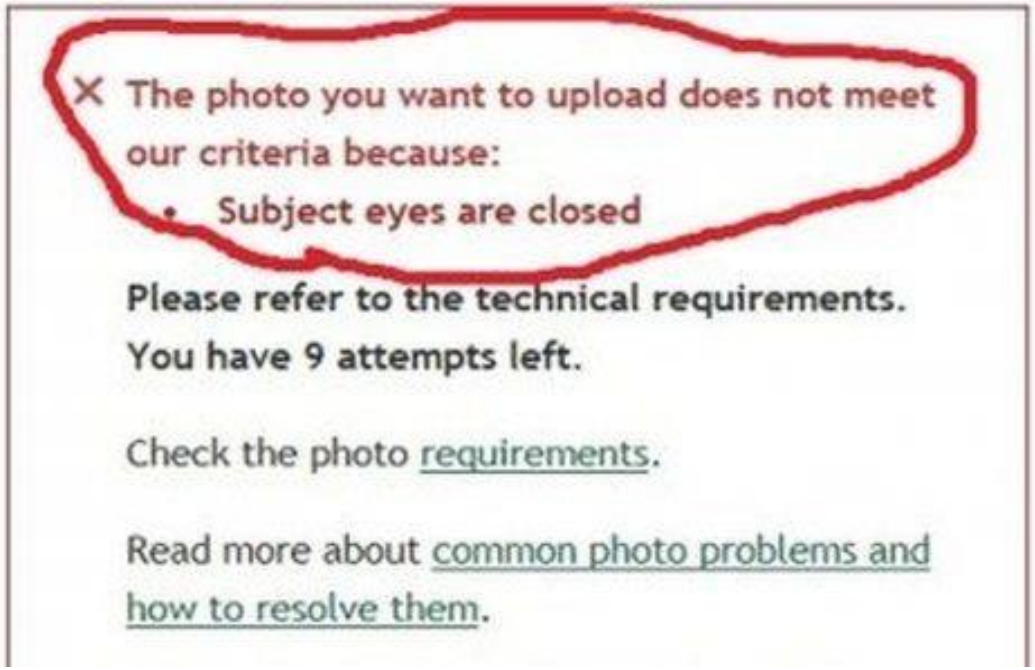
1. Risks of AI/ML in real world
2. AI perceptions, explainability and interpretability
3. **Investigating Bias**
4. Ameliorating bias

## Investigating Bias: What's the Problem?

- An ML Failure occurs when a ML system does something **unanticipated** and/or **undesirable**.

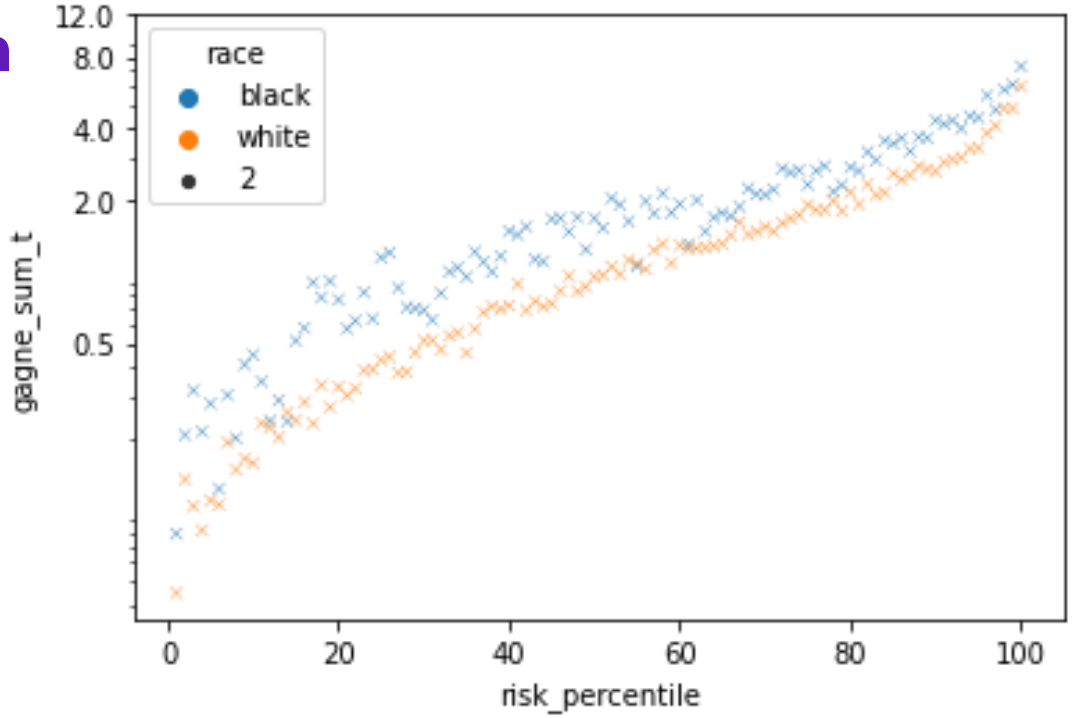
## … in Facial Recognition,

**an ML Failure… makes it hard for someone of East Asian descent to get their passport.**



# … in Healthcare,

**an ML Failure leads to… worse medical care for Black patients than for white patients.**



# Going Back to Our Definition ...

"An ML Failure occurs when a ML system does something **unanticipated** and/or **undesirable**."

## Undesirable to whom?

- Designers?
- Users?
- Shareholders?

## Unanticipated by whom?

- Designers?
- Users?
- Shareholders?

# ML bias is a sociotechnical problem.

Technical approaches are **remediations** –
not solutions.

We are learning how to **understand** bias, and putting some
**tools in our toolbox** to make bias less harmful.

# Bias Vs Fairness

**Bias**: Inclination or prejudice for or against one person or group, especially in a way considered to be unfair

An algorithm is **fair** if it makes predictions that do not favour or discriminate against certain individuals or groups based on sensitive characteristics..

# Race-conscious admissions

**Katharine Meyer**
June 29, 2023

- → The Supreme Court ruled current race-based affirmative action policies violate the Equal Protection Clause of the 14th Amendment; colleges will need to make immediate changes to their existing application review processes.
- → Insights from state bans on race-conscious admissions suggest Black and Hispanic enrollment could drop 10% at selective colleges nationwide.
- → As colleges try to maintain a commitment to institutional diversity, they should eliminate policies such as legacy admissions and early decisions that advantage the most advantaged.
- → While much remains to be understood about the *SFFA* decision and its impact, one clear takeaway is that this will not be the last word on how race can factor into college admissions.

## Investigating Bias: Why is it a big deal?

- There are **protected attributes** that algorithms impede upon.

# Isn’t the point of ML to discriminate?

**Yes, to a point.** In ML bias, we are interested in *preventing* discrimination based on “socially salient qualities that have **served as the basis for unjustified and systematically adverse treatment** in the past.”

(Hardt, 2017)

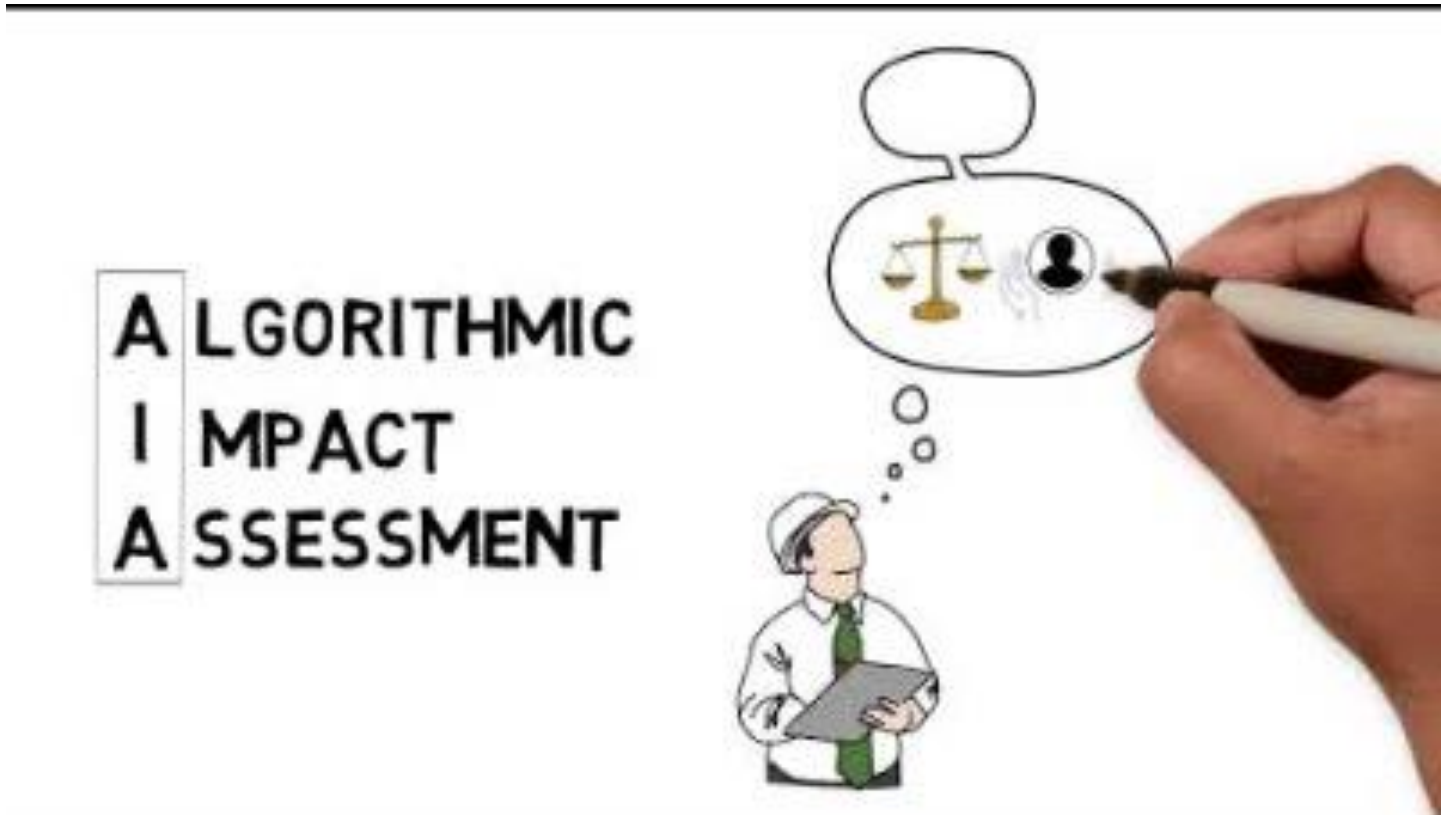
# Publicly Regulated Domains

## There's legal precedent that protects people from discrimination in various activities.

- Credit (Equal Credit Opportunity Act)
- Education (Civil Rights Act of 1964; Education Amendments of 1972)
- Employment (Civil Rights Act of 1964)
- Housing (Fair Housing Act)
- 'Public Accommodation' (Civil Rights Act of 1964)
- Race; Color; Religion; National origin (Civil Rights Act of 1964)
- Citizenship; Age; Sex; Familial status; Pregnancy; Disability status; Veteran status; Genetic Information

# Example: Canada's AI Policies

**Different countries look at discrimination differently. Canada treats AI differently!**



# Deep Dive into the Civil Rights Act

**There's legal precedent for evaluating whether treatment is biased in law. We'll see later how these ideas apply in machine learning context!**

- Disparate treatment
  - Formal: Explicitly considering class membership, even if it's relevant
  - Informal: Purposefully attempting to discriminate without direct reference to class membership
- Disparate impact
  - Four-fifths rule: Selection rate for a certain group is less than 80% of rate for group with highest selection rate, there is adverse impact.
  - Alternative practice: Could the same goal be achieved using a different procedure that would result in a smaller disparity?

# Beyond the law: bias in images

# Thinking Beyond the Law

**We don’t just combat ML failures with the law. People are combating bias through many different means.**

- Worker organizing
  - Google, Alphabet Workers Union
- Design & business decisions
  - Office of Responsible AI @ Microsoft
- Moral imperatives
- Equity, inclusion, belonging
- How can we do better?

## Investigating Bias: Why does it happen?

- Bias can occur at **all points in the cycle** by various different stakeholders.

# Types of Errors

## Imbalanced Data Sets

→ Amazon's hiring model
→ Google's CV model

## Removing Sensitive Features

→ If a credit score model 'removes' race, the model will learn racial discrimination

## Lurking Bias in Features

→ Facial recognition based on white faces
→ Credit score & race

## Poorly Framed Problems

→ Predicting if someone is a criminal based on their face

# Errors Lead to …

# … Wrongful Convictions

# Call to Action!

| | |
|---|---|
| Designers aren't trained to look. | Be aware of & identify biases in an algorithm before it launches. |
| Users are too trusting of systems. | Stay vigilant and assess algorithms' fairness, and teach others to do the same. |
| Watchdogs and protection agencies are overloaded. | Join an organization, write articles, or educate the public. |
| Fairness is itself a contested concept. | Understand that fairness varies socially, culturally, and contextually. |

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4. **Ameliorating bias**

# Ameliorating Bias: What's the Lingo?

Let's demystify some of the **jargon** that experts use to talk about ML bias.

# Grouping Terminology

## Privileged Group

We expect this group to get the favorable outcome **more often** than they should.

## Unprivileged Group

We expect this group to get the favorable outcome **less often** than they should.

| | | Privileged Group | Unprivileged Group |
|---|---|---|---|
| Adult Census Income | Race | White | Non-White |
| | Sex | Male | Non-Male |
| Recidivism (Compas) | Race | White | Non-White |
| | Sex | Female | Male |

# A note on the formalism

## Privilege

Systemic inequality and disparate treatment resulting from societal differences in which demographic groups hold power

## "Privileged group"

A mathematical formalism describing a group (any group) that gets a favorable outcome (any outcome) more often than they "should"

**MIND THE GAP!**

→ Privilege is what we actually care about

→ The formality of "privileged groups" can help us understand privilege - but the two are ***not*** the same.

Thanks to Lauren Chambers

# Ameliorating Bias: How can we find bias?

There are **concrete strategies to quantify bias** and assess its severity.

# Review: Why can't we be blind to bias?

**Always include the sensitive features (like race and gender)**

→ Sensitive features cannot be ignored.

→ **If algorithms "hear no evil", we can "see no evil"**: "non-sensitive" features in our dataset may be correlated with sensitive features; if we remove those features, we will not know our algorithms exhibit bias.

# Strategies to Identify Bias

**DISPARATE IMPACT**

STATISTICAL PARITY

EQUAL OPPORTUNITY

AVERAGE ODDS

**Each group should have an *equal opportunity* of achieving the favorable outcome.**
We calculate the ratio of rate of favorable outcome for unprivileged group compared to that of privileged group.

**The ideal value is 1.**
A value < 1 implies there is benefit toward the privileged group.

| | Trait 1 | Privileged | Ratio | Trait 2 | Privileged | Ratio |
|---|---|---|---|---|---|---|
| Adult Income | Race | White | 0.55 | Sex | Male | 0.29 |
| Recidivism (Compas) | Race | White | 0.75 | Sex | Female | 0.59 |

How does disparate impact compare to the 4/5 rule?

# Strategies to Identify Bias

DISPARATE IMPACT

- STATISTICAL PARITY
- EQUAL OPPORTUNITY
- AVERAGE ODDS

**Statistical Parity:**
Demographics of those receiving any classification should be the same as demographics of the underlying population.

**Equal Opportunity / Average Odds**
Each group should be classified (in)correctly at the same rate.

# Strategies to Identify Bias

DISPARATE IMPACT

**STATISTICAL PARITY**

EQUAL OPPORTUNITY

AVERAGE ODDS

**Demographics of those receiving any classification should be the same as demographics of the underlying population.**
We take the difference of rate of favorable outcomes by rate of favorable outcomes by unprivileged group.

**The ideal value is 0.**
A value < 0 implies there is benefit toward the privileged group.

| | Trait 1 | Privileged | Ratio | Trait 2 | Privileged | Ratio |
|---|---|---|---|---|---|---|
| Adult Income | Race | White | -0.18 | Sex | Male | -0.33 |
| Recidivism (Compas) | Race | White | -0.18 | Sex | Female | -0.36 |

When might statistical parity and disparate impact disagree?

# Strategies to Identify Bias

DISPARATE IMPACT

STATISTICAL PARITY

**EQUAL OPPORTUNITY**

AVERAGE ODDS

**Each group should be 'equally' incorrectly classified.**
We take the difference of true positive rates between unprivileged and privileged groups.

**The ideal value is 0.**
A value < 0 implies there is benefit toward the privileged group.

| | Trait 1 | Privileged | Ratio | Trait 2 | Privileged | Ratio |
|---|---|---|---|---|---|---|
| Adult Income | Race | White | -0.06 | Sex | Male | -0.14 |
| Recidivism (Compas) | Race | White | -0.12 | Sex | Female | -0.30 |

Where is there the **most bias** in these 2 datasets?

# Strategies to Identify Bias

DISPARATE IMPACT

STATISTICAL PARITY

EQUAL OPPORTUNITY

**AVERAGE ODDS**

**Each group should be 'equally' incorrectly classified.**
We take the average difference of false positive rate and true positive rate between unprivileged and privileged groups.

**The ideal value is 0.**
A value < 0 implies there is benefit toward the privileged group.

| | Trait 1 | Privileged | Ratio | Trait 2 | Privileged | Ratio |
|---|---|---|---|---|---|---|
| Adult Income | Race | White | -0.09 | Sex | Male | -0.19 |
| Recidivism (Compas) | Race | White | -0.16 | Sex | Female | -0.35 |

Where is there the **most bias** in these 2 datasets?

## Ameliorating Bias: How do we make it better?

While there is no way to “fix” bias, **there are methods for making bias less harmful**.

# Misconceptions

## Bias Starts in Data

Bias can start **anywhere**: pre-processing, post-processing, domain specification, models...

## Algorithms Don't Create Bias – They Transmit it

If an algorithm can be de-biased with a dataset, they can themselves be biased.

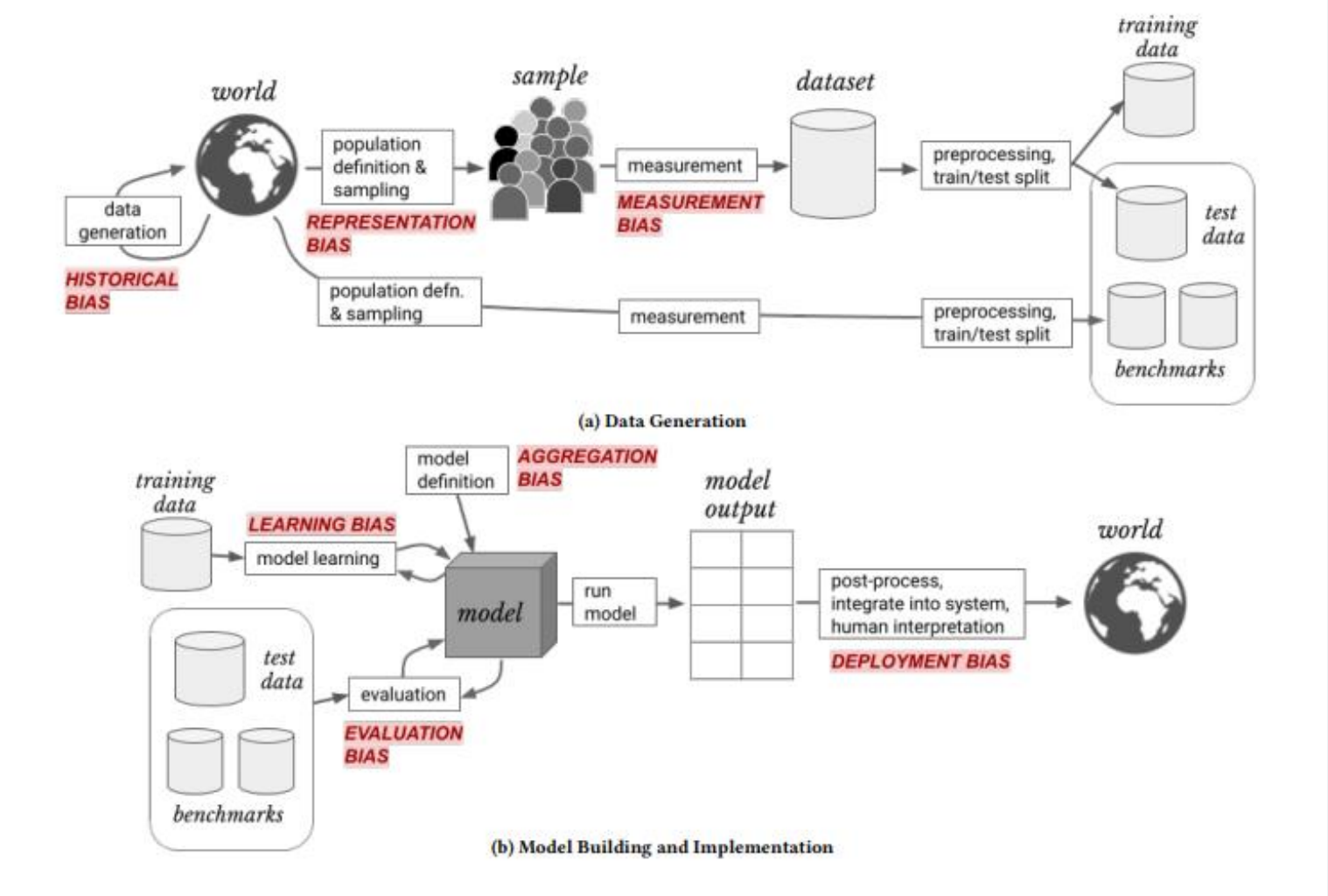
## Race & gender are the most obvious biases.

**These are the least obvious**. they're protected: least likely included in training data.

## Bias is Fixable

"All data embeds a worldview & **all models have some bias**. Most interventions just try to make the model biased towards more inclusive (& non-illegal) outcomes."

# Bias can start anywhere




Suresh & Guttag, 2021.

# Strategies to Mitigate Bias

- **CONSTRAINTS**
- REWEIGHTING
- OPTIMIZED PRE-PROCESSING
- ADVERSARIAL DEBIASING
- REJECT-OPTION-BASED CLASSIFICATION

**Fairness constraints allow us to specify a tradeoff between a classifier's "fairness" and its accuracy.**
Sometimes, our dataset is badly biased. For example, a dataset of past hiring decisions may embed a bias against women. In this case, an "accurate" classifier would be unfair - perhaps illegally so.

**To correct for this, we can set a fairness constraint (e.g., a minimum disparate impact score).**
With this constraint, the classifier will be as accurate as possible while exceeding the minimum disparate impact score.

# Strategies to Mitigate Bias

- **REWEIGHTING**
- OPTIMIZED PRE-PROCESSING
- ADVERSARIAL DEBIASING
- REJECT-OPTION-BASED CLASSIFICATION

Weights the examples in each (group, label) combination differently to ensure fairness before classification.

# Strategies to Mitigate Bias

- REWEIGHTING
- **OPTIMIZED PRE-PROCESSING**
- ADVERSARIAL DEBIASING
- REJECT-OPTION-BASED CLASSIFICATION

Learns a probabilistic transformation that can modify the features and the labels in the training data.

# Strategies to Mitigate Bias

REWEIGHTING

OPTIMIZED PRE-PROCESSING

**ADVERSARIAL DEBIASING**

REJECT-OPTION-BASED CLASSIFICATION

Learns a classifier that maximizes prediction accuracy and simultaneously reduces an adversary's ability to determine the protected attribute from the predictions.

Since the predictions cannot carry any group discrimination information that the adversary can exploit, the classifier must be fair (right?).

# Strategies to Mitigate Bias

| | |
|---|---|
| REWEIGHTING | Changes predictions from a classifier to make them fairer. |
| OPTIMIZED PRE-PROCESSING | Provides favorable outcomes to unprivileged groups and unfavorable outcomes to privileged groups in a confidence band around the decision boundary with the highest uncertainty. |
| ADVERSARIAL DEBIASING | |
| **REJECT-OPTION-BASED CLASSIFICATION** | |

# Summary

- ML bias is **sociotechnical**
  - It is not **caused** by technical problems alone, and cannot be **“solved”** by technical solutions alone

- **Technical approaches are one** way of understanding and beginning to address ML bias - a tool in the toolbox
  - Detecting bias
  - Ameliorating bias

# What's Next?

There's **a lot more work to do** in this space – here is a non-comprehensive list of what can come after today.

## Accountability

Assigning responsibility for harm.

## Transparency & Explainability

How do algorithms make the decisions they make?

## Environmental Consequences

Training large models means large energy requirements.

## Reinforcement Learning

Reward function specification, existential risk/x-risk

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5. **Evaluation on Biased Dataset**

# Take Multi-Way Classification as one Example

**Binary classification** $Y = \{0,1\}$

**Multi-way classification**

- **K classes** $Y = \{0,1,\ldots,K-1\}$
- **Goal:** learn to classify correctly K classes
- Or **learn** $f : X \rightarrow \{0,1,\ldots,K-1\}$

**Errors:**

- **Zero-one (misclassification) error for an example:**

$$Error_1(\mathbf{x}_i, y_i) = \begin{cases} 1 & f(\mathbf{x}_i, \mathbf{w}) \neq y_i \\ 0 & f(\mathbf{x}_i, \mathbf{w}) = y_i \end{cases}$$

- **Mean misclassification error (for a dataset):**

$$\frac{1}{n}\sum_{i=1}^{n} Error_1(\mathbf{x}_i, y_i)$$

# Long-Tail Distribution

Imbalance degree: max/min

# Long-Tail at Evaluation

| Class Index | Number of Test Samples | Correctly Recognized Samples | Accuracy |
|---|---|---|---|
| 1 | 100 | 99 | 0.99 |
| 2 | 50 | 30 | 0.6 |
| 3 | 20 | 5 | 0.25 |
| 4 | 10 | 1 | 0.1 |
| 5 | 5 | 0 | 0 |

Accuracy: 135/185 ≈ 73.0% or 38.8%

Take-Home Question: When the training samples exhibit long-tail distribution but the test data per class are similar, how should we examine the accuracy on test set.